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ELECTRONIC DEVICE

Abstract

An electronic device and an operating method thereof are provided. The electronic device includes a driving transistor and a preset transistor. The operating method includes the following steps: turning on the preset transistor, and providing a first voltage to a control terminal of the driving transistor; turning off the preset transistor, and providing a second voltage to the driving transistor, wherein the first voltage is greater than the second voltage; executing a sensing process; and executing a scan process to provide an output signal.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation application of and claims the priority benefit of a prior U.S. application Ser. No. 18/321,795, filed on May 23, 2023. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

[0002] The disclosure relates a device; particularly, the disclosure relates to an electronic device and an operating method thereof.

Description of Related Art

[0003] For a general active pixel sensor (APS), since the active pixel sensor has the threshold voltage (V_{th}) problem of the driving transistor, such as in-plane V_{th} variation due to process and V_{th} degradation caused by various factors (e.g. static destruction, irradiation etc.), which greatly affects the sensing quality of image sensing results.

SUMMARY

[0004] The operating method of the disclosure is adapted to the electronic device. The electronic device includes a driving transistor and a preset transistor. The operating method includes the following step: turning on the preset transistor, and providing a first voltage to a control terminal of the driving transistor; turning off the preset transistor, and providing a second voltage to the driving transistor, wherein the first voltage is greater than the second voltage; executing a sensing process; and executing a scan process to provide an output signal.

[0005] The electronic device of the disclosure includes a preset transistor and a driving transistor. The driving transistor is electrically connected to the preset transistor. A first voltage is provided to a control terminal of the driving transistor when the preset transistor is turned on. A second voltage is provided to the driving transistor when the preset transistor is turned off. The first voltage is greater than the second voltage.

[0006] Based on the above, according to the operating method and the electronic device of the disclosure, the electronic device can provide a good image sensing function.

[0007] To make the aforementioned more comprehensible, several embodiments accompanied with drawings are described in detail as follows.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The accompanying drawings are included to provide a further understanding of the disclosure, and are incorporated in and constitute a part of this specification. The drawings illustrate exemplary embodiments of the disclosure and, together with the description, serve to explain the principles of the disclosure.

[0009] FIG. 1 is a schematic diagram of an electronic device according to an embodiment of the disclosure.

[0010] FIG. 2 is a schematic diagram of a pixel circuit and a readout circuit according to an

embodiment of the disclosure.

[0011] FIG. 3 is a schematic diagram of voltage variation according to the embodiment of FIG. 2 of the disclosure.

[0012] FIG. 4 is a flowchart of an operating method according to an embodiment of the disclosure.

[0013] FIG. 5 is a schematic diagram of an electronic device according to another embodiment of the disclosure.

[0014] FIG. 6 is a schematic diagram of a pixel circuit and a readout circuit according to another embodiment of the disclosure.

[0015] FIG. 7 is a schematic diagram of voltage variation according to the embodiment of FIG. 6 of the disclosure.

DESCRIPTION OF THE EMBODIMENTS

[0016] Reference will now be made in detail to the exemplary embodiments of the disclosure, examples of which are illustrated in the accompanying drawings. Whenever possible, the same reference numbers are used in the drawings and the description to refer to the same or like components.

[0017] Certain terms are used throughout the specification and appended claims of the disclosure to refer to specific components. Those skilled in the art should understand that electronic device manufacturers may refer to the same components by different names. This article does not intend to distinguish those components with the same function but different names. In the following description and rights request, the words such as “comprise” and “include” are open-ended terms, and should be explained as “including but not limited to . . .”.

[0018] The term “coupling (or connection)” used throughout the whole specification of the present application (including the appended claims) may refer to any direct or indirect connection means. For example, if the text describes that a first device is coupled (or connected) to a second device, it should be interpreted that the first device may be directly connected to the second device, or the first device may be indirectly connected through other devices or certain connection means to be connected to the second device. The terms “first”, “second”, and similar terms mentioned throughout the whole specification of the present application (including the appended claims) are merely used to name discrete elements or to differentiate among different embodiments or ranges. Therefore, the terms should not be regarded as limiting an upper limit or a lower limit of the quantity of the elements and should not be used to limit the arrangement sequence of elements. In addition, wherever possible, elements/components/steps using the same reference numerals in the drawings and the embodiments represent the same or similar parts. Reference may be mutually made to related descriptions of elements/components/steps using the same reference numerals or using the same terms in different embodiments.

[0019] FIG. 1 is a schematic diagram of an electronic device according to an embodiment of the disclosure. Referring to FIG. 1, the electronic device **100** may be an active pixel sensor (APS), and may be manufactured by thin-film transistor (TFT) process, but the disclosure is not limited thereto. The electronic device **100** includes an active area and a peripheral area, a (active matrix (AM)) pixel array **110** is located in the active area, and a peripheral circuit **120** is located in the peripheral area. The pixel array **110** is electrically connected to the peripheral circuit **120**. The pixel array **110** includes a plurality of pixel units $p(1,1)$ to $p(N,M)$, a plurality of reset signal lines **101_1** to **101_N**, a plurality of read signal lines **102_1** to **102_N** and a plurality of data signal lines **103_1** to **103_M**, wherein M and N are positive integers. The peripheral circuit **120** includes a plurality of readout circuits **121_1** to **121_M**, a preset signal line **104**, a preset voltage line **105** and a readout control line **106**. In the embodiment of the disclosure, one row of the pixel units $p(1,1)$ to $p(N,M)$ is electrically connected to corresponding one of the reset signal lines **101_1** to **101_N** and corresponding one of the read signal lines **102_1** to **102_N**. One column of the pixel units $p(1,1)$ to $p(N,M)$ is electrically connected to corresponding one of the readout circuits **121_1** to **121_M** through a corresponding one of the data signal lines **103_1** to **103_M**. In the embodiment of the

disclosure, the readout circuits **121_1** to **121_M** are electrically connected to the preset signal line **104**, the preset voltage line **105** and the readout control line **106**.

[0020] FIG. 2 is a schematic diagram of a pixel circuit and a readout circuit according to an embodiment of the disclosure. Referring to FIG. 1 and FIG. 2, the specific circuit architecture of at least one of the pixel units $p(1,1)$ to $p(N,M)$ and corresponding one of the readout circuits **121_1** to **121_M** of FIG. 1 may be implemented as the pixel unit **211** and readout circuit **221** of FIG. 2. The pixel unit **211** may be a pixel at n -th row and m -th column in the pixel array **110**. In the embodiment of the disclosure, the pixel unit **211** includes a read transistor **M1**, a driving transistor **M2**, a reset transistor **M3** and a sensing unit **PD**. A first terminal of the read transistor **M1** is electrically connected to the driving transistor **M2** and the reset transistor **M3** through a circuit node **P**. A second terminal of the read transistor **M1** is electrically connected to the readout circuit **221** through a circuit node **N2** and the data signal line **103_m**, wherein m is an integer between 1 and M . A control terminal of the read transistor **M1** is configured to receive a read signal **RD(n)** from the read signal line **102_n**, wherein n is an integer between 1 and N . A first terminal of the driving transistor **M2** is configured to receive an adjustable voltage V_a . A second terminal of the driving transistor **M2** is electrically connected to the read transistor **M1** and the reset transistor **M3** through the circuit node **P**. A control terminal of the driving transistor **M2** is electrically connected to the sensing unit **PD** and the reset transistor **M3** through a circuit node **N1**. A first terminal of the reset transistor **M3** is electrically connected to the read transistor **M1** and the driving transistor **M2** through the circuit node **P**. A second terminal of the reset transistor **M3** is electrically connected to the sensing unit **PD** and the driving transistor **M2** through the circuit node **N1**. To be more specific, reset transistor **M3** is electrically connected between the second terminal and the control terminal of the driving transistor **M2**. A control terminal of the reset transistor **M3** is configured to receive a reset signal **RS(n)** from the reset signal line **101_n**. An anode of the sensing unit **PD** is electrically connected to a bias voltage V_{bias} . A cathode of the sensing unit **PD** is electrically connected to the driving transistor **M2** and the reset transistor **M3** through the circuit node **N1**.

[0021] In the embodiment of the disclosure, the peripheral circuit **120** includes a preset transistor **Tprst**, a readout transistor **Tro**, an amplifier circuit **221_1** and a current source **221_2**. The preset transistor **Tprst** is included in the peripheral circuit **120**, but the disclosure is not limited thereto. A first terminal of the preset transistor **Tprst** is configured to receive a preset voltage V_{prst} from the preset voltage line **105**. A second terminal of the preset transistor **Tprst** is electrically connected to the circuit node **N2** and the readout transistor **Tro**. A control terminal of the preset transistor **Tprst** is configured to receive a preset signal **Prst** from the preset signal line **104**. A first terminal of the readout transistor **Tro** is electrically connected to the circuit node **N2** and the preset transistor **Tprst**. A second terminal of the readout transistor **Tro** is electrically connected to the amplifier circuit **221_1** and the current source **221_2** through a circuit node **N3**. A control terminal of the readout transistor **Tro** is configured to receive a readout signal **RO** from the readout control line **106**. An input terminal of the amplifier circuit **221_1** is electrically connected to the readout transistor **Tro** and the current source **221_2** through the circuit node **N3**. The current source **221_2** is electrically connected between the circuit node **N3** and a ground or a power supply, and configured to generate a constant current I_{cs} flowing from the circuit node **N3** toward the ground or the power supply.

[0022] In the embodiment of the disclosure, the read transistor **M1**, the driving transistor **M2**, the reset transistor **M3**, the preset transistor **Tprst** and the readout transistor **Tro** may be N-type transistors, but the disclosure is not limited thereto. In the embodiment of the disclosure, the first terminal and the second terminal of one of the above transistors may be a drain terminal and a source terminal respectively, and the control terminal of the transistor may be a gate terminal, but the disclosure is not limited thereto. In one embodiment of the disclosure, at least one of the read transistor **M1**, the driving transistor **M2**, the reset transistor **M3**, the preset transistor **Tprst** and the readout transistor **Tro** may be a P-type transistor. In the embodiment of the disclosure, the sensing unit **PD** may be a photodiode, but the disclosure is also not limited thereto.

[0023] FIG. 3 is a schematic diagram of voltage variation according to the embodiment of FIG. 2 of the disclosure. FIG. 4 is a flowchart of an operating method according to an embodiment of the disclosure. Referring to FIG. 1 to FIG. 4, the pixel unit 211 and the readout circuit 221 of FIG. 2 may perform the following steps S410 to S440. In step S410, the preset transistor Tprst may be turned on, and a first voltage (voltage V.sub.BE of the circuit node N1) may be provided to the control terminal of the driving transistor M2. As shown in FIG. 3, the pixel unit 211 may operate a reset process during a reset period RP from time t1 to time t4, and the reset process includes a preset process during the period from time t1 to time t2. During the period from time t0 to time t3, the adjustable voltage Va may be changed from a reset voltage Vrst to an operation voltage Vdd. However, in one embodiment of the disclosure, the first terminal of the driving transistor M2 may also be changed to a floating state (i.e. high impedance). During the period from time t1 to time t2, the reset signal RS(n), the preset signal Prst and the read signal RD(n) may be changed from a low voltage level to a high voltage level respectively, and the readout signal RO may be maintained at the low voltage level. Thus, the read transistor M1, the reset transistor M3 and the preset transistor Tprst may be turned on, and the readout transistor Tro may be turned off, and the preset voltage Vprst (i.e. the first value) may be provided to the control terminal of the driving transistor M2. In other words, the voltage value of the first voltage (i.e. voltage V.sub.BE of the circuit node N1) is at a first value equal to the preset voltage Vprst. The preset transistor Tprst is turned on during at least a part of the reset process. It should be noted that in FIG. 3, both the preset transistor Tprst and the read transistor M1 are turned on during a part of the reset process, but since the preset voltage Vprst may be provided to the control terminal of the driving transistor M2 when the preset transistor Tprst, the read transistor M1 and the reset transistor M3 are all turned on, it may be that only one of the preset transistor Tprst and the read transistor M1 is turned on during a part of the reset process, another one of them is turned on during all the reset process. On the other hand, the control terminal and the second terminal of the driving transistor M2 may be short-circuited by the turned on reset transistor M3, since the driving transistor M2 has a symmetrical structure in principle, the driving transistor M2 can be regarded as an equivalent diode with exchanged source and drain. A cathode terminal of the equivalent diode corresponds to the first terminal of the driving transistor M2. An anode terminal of the equivalent diode corresponds to the control terminal and the second terminal of the driving transistor M2. Thus, due to the operation voltage Vdd is greater than the preset voltage Vprst, the equivalent diode is operated in reverse bias, and there is no current flowing through the equivalent diode. Therefore, a voltage V.sub.BE (i.e. the first voltage) of the circuit node N1 is equal to the preset voltage Vprst during the period from time t1 to time t3. It should be noted that since the transistors in the disclosure are N-type transistors, the term “low voltage level” in the disclosure is referred to a voltage level to turn off the N-type transistor (for example, a ground), and the term “high voltage level” in the disclosure is referred to a voltage level high enough to turn on the N-type transistor.

[0024] In step S420, the preset transistor Tprst may be turned off, and a second voltage may be provided to a first terminal of the driving transistor M2. As shown in FIG. 3, during the period from time t3 to time t4, the adjustable voltage Va may be changed from the operation voltage Vdd to the reset voltage Vrst. During the period from time t2 to time t4, the preset signal Prst and the read signal RD(n) may be changed from the high voltage level to the low voltage level respectively, the reset signal RS(n) may be maintained at the high voltage level, and the readout signal RO may be maintained at the low voltage level. Thus, the reset transistor M3 may be turned on, and the read transistor M1, the preset transistor Tprst and the readout transistor Tro may be turned off, and the reset voltage Vrst (i.e. the second voltage) may be provided to the first terminal of the driving transistor M2. More specifically, the control terminal and the second terminal of the driving transistor M2 may also be short-circuited by the turned on reset transistor M3, and the driving transistor M2 may also be operated as the equivalent diode. Thus, due to the reset voltage Vrst is smaller than the preset voltage Vprst, the equivalent diode is operated in forward bias, and there is

a diode current flowing through the equivalent diode. Therefore, the voltage $V_{\text{sub.BE}}$ (i.e. the first voltage) of the circuit node N1 is pulled down after time t_3 , and after certain time, the voltage $V_{\text{sub.BE}}$ (i.e. the first voltage) of the circuit node N1 is equal to the voltage (i.e. the second value) of the reset voltage V_{rst} plus a threshold voltage V_{th} (i.e. $V_{\text{rst}}+V_{\text{th}}$) of the driving transistor M2. That is, the voltage $V_{\text{sub.BE}}$ (i.e. the first voltage) of the circuit node N1 is reduced from the preset voltage V_{prst} (i.e. the first value) to the voltage (i.e. the second value) of the reset voltage V_{rst} plus a threshold voltage V_{th} (i.e. $V_{\text{rst}}+V_{\text{th}}$) when the reset voltage V_{rst} is provided to the driving transistor M2. Therefore, the second value of the first voltage ($V_{\text{rst}}+V_{\text{th}}$) is greater than a voltage value of the second voltage (V_{rst}).

[0025] Moreover, during the period from time t_1 to time t_5 , the voltage $V_{\text{data(m)}}$ of the circuit node N2 is equal to the preset voltage V_{prst} . Then, during a data initialization period DIP from time t_5 to time t_6 , the electronic device 100 may operate a data initialization process. During the data initialization period DIP from time t_5 to time t_6 , the reset signal $\text{RS}(n)$, the preset signal Prst and the read signal $\text{RD}(n)$ may be maintained at the low voltage level respectively, and the readout signal RO may be changed from the low voltage level to the high voltage level. Thus, the read transistor M1, the reset transistor M3 and the preset transistor Tprst may be turned off, and the readout transistor Tro may be turned on, and the voltage $V_{\text{data(m)}}$ of the circuit node N2 is pulled down from the preset voltage V_{prst} to a low voltage level (i.e. ground voltage).

[0026] In step S430, the pixel unit 211 executes a sensing process. As shown in FIG. 3, the electronic device 100 may operate the sensing process during an exposure period EP from time t_7 to time t_8 , and the sensing unit PD may provide a sensing signal (sensing current) corresponding to a sensing result to the control terminal of the driving transistor M2. During the exposure period EP from time t_7 to time t_8 , the reset signal $\text{RS}(n)$, the preset signal Prst , the read signal $\text{RD}(n)$ and the readout signal RO may be maintained at the low voltage level. Thus, the voltage $V_{\text{sub.BE}}$ (i.e. the first voltage) of the circuit node N1 is further reduced from the voltage (i.e. the second value) of the reset voltage V_{rst} plus a threshold voltage V_{th} (i.e. $V_{\text{rst}}+V_{\text{th}}$) to the voltage (i.e. the third value) of the reset voltage V_{rst} minus a sensing voltage $\Delta V_{\text{sub.BE}}$ and plus a threshold voltage V_{th} (i.e. $V_{\text{rst}}-\Delta V_{\text{sub.BE}}+V_{\text{th}}$). The sensing voltage/ $V_{\text{sub.BE}}$ represents a voltage variation due to the sensing result of the sensing unit PD.

[0027] In step S440, the pixel unit 211 executes a scan process to provide an output signal. As shown in FIG. 3, the pixel unit 211 may operate a scan process during a scan period SP from time t_9 to time t_{12} . During the scan period SP from time t_9 to time t_{12} , the adjustable voltage V_a may be changed from the reset voltage V_{rst} to the operation voltage V_{dd} . During a readout period ROP from time t_{10} to time t_{11} , the read signal $\text{RD}(n)$ and the readout signal RO may be changed from the low voltage level to the high voltage level respectively, and the reset signal $\text{RS}(n)$ and the preset signal Prst may be maintained at the low voltage level. Thus, the read transistor M1 and the readout transistor Tro may be turned on, and the reset transistor M3 and the preset transistor Tprst may be turned off, and the readout circuit 221 may readout the output signal with the voltage $V_{\text{data(m)}}$ through the circuit node N2 according to the voltage $V_{\text{sub.BE}}$ (i.e. the first voltage) of the circuit node N1. More specifically, the constant current I_{cs} is flowing through the driving transistor M2, the read transistor M1 and the readout transistor Tro and toward the current source 221_2. The driving transistor M2 may act as a source follower in the saturation region, and the voltage $V_{\text{data(m)}}$ of the circuit node N2 is pulled up from the low voltage level to the voltage of the reset voltage V_{rst} minus the sensing voltage/ $V_{\text{sub.BE}}$ and minus a transistor voltage β (i.e.

$(V_{\text{rst}}-\Delta V_{\text{sub.BE}}+V_{\text{th}})-V_{\text{th}}-\beta=V_{\text{rst}}-\Delta V_{\text{sub.BE}}-\delta$). That is, the threshold voltage V_{th} can be excluded in the voltage $V_{\text{data(m)}}$. The transistor voltage β may be expressed in the following equation (1), wherein the symbol L represents a gate width of the driving transistor M2, the symbol W represents a channel width of the driving transistor M2, the symbol μ represents an electron mobility and the symbol C_{ox} represents a capacitance per unit area of the gate insulator of the driving transistor M2.

$$[00001] \quad = \sqrt{\frac{2 \cdot \text{Math. } I_{CS} \cdot \text{Math. } L}{W \cdot \text{Math. } \cdot \text{Math. } Cox}} \quad \text{equation}(1)$$

[0028] Therefore, the input terminal of the amplifier circuit **221_1** may receive an input voltage V_{in} from the circuit node **N3**, and the input voltage V_{in} is equal to the voltage $V_{data(m)}$ (i.e. $V_{rst} - \Delta V_{sub} - \beta$). That is, the readout circuit **221** may generate the sensing data that is not affected by the threshold voltage V_{th} of the driving transistor **M2**.

[0029] FIG. 5 is a schematic diagram of an electronic device according to another embodiment of the disclosure. Referring to FIG. 5, the electronic device **500** may be an APS, and includes a (active matrix (AM)) pixel array **510** and a peripheral circuit **520**. The pixel array **510** is electrically connected to the peripheral circuit **520**. The pixel array **510** includes a plurality of pixel units $p(1,1)$ to $p(N,M)$, a plurality of reset signal lines **501_1** to **501_N**, a plurality of read signal lines **502_1** to **502_N**, a plurality of preset signal lines **503_1** to **503_N** and a plurality of data signal lines **504_1** to **504_M**, wherein M and N are positive integers. The peripheral circuit **520** includes a plurality of readout circuits **521_1** to **521_M**, a reset control line **505**, a reset voltage line **506** and a readout control line **507**. In the embodiment of the disclosure, one row of the pixel units $p(1,1)$ to $p(N,M)$ is electrically connected to corresponding one of the reset signal lines **501_1** to **501_N**, corresponding one of the read signal lines **502_1** to **502_N** and corresponding one of the preset signal lines **503_1** to **503_N**. One column of the pixel units $p(1,1)$ to $p(N,M)$ is electrically connected to corresponding one of the readout circuits **521_1** to **521_M** through a corresponding one of the data signal lines **504_1** to **504_M**. In the embodiment of the disclosure, the readout circuits **521_1** to **521_M** are electrically connected to the reset control line **505**, the reset voltage line **506** and the readout control line **507**.

[0030] FIG. 6 is a schematic diagram of a pixel circuit and a readout circuit according to another embodiment of the disclosure. Referring to FIG. 5 and FIG. 6, the specific circuit architecture of at least one of the pixel units $p(1,1)$ to $p(N,M)$ and corresponding one of the readout circuits **521_1** to **521_M** of FIG. 5 may be implemented as the pixel unit **611** and readout circuit **621** of FIG. 6. The pixel unit **611** may be a pixel at n -th row and m -th column in the pixel array **510**. In the embodiment of the disclosure, the pixel unit **611** includes a read transistor **M1**, a driving transistor **M2**, a reset transistor **M3**, a preset transistor **M4** and a sensing unit **PD**. A first terminal of the read transistor **M1** is electrically connected to the driving transistor **M2** through a circuit node **P**. A second terminal of the read transistor **M1** is electrically connected to the readout circuit **621** through a circuit node **N2** and the data signal line **504_m**, wherein m is an integer between 1 and M . A control terminal of the read transistor **M1** is configured to receive a read signal $RD(n)$ from the read signal line **502_n**, wherein n is an integer between 1 and N . A first terminal of the driving transistor **M2** is electrically connected to the preset transistor **M4** and the reset transistor **M3**. A second terminal of the driving transistor **M2** is electrically connected to the read transistor **M1** through the circuit node **P**. A control terminal of the driving transistor **M2** is electrically connected to the sensing unit **PD** and the reset transistor **M3** through a circuit node **N1**. A first terminal of the reset transistor **M3** is electrically connected to the sensing unit **PD** and the driving transistor **M2** through the circuit node **N1**. A second terminal of the reset transistor **M3** is electrically connected to the driving transistor **M2** and the preset transistor **M4**. A control terminal of the reset transistor **M3** is configured to receive a reset signal $RS(n)$ from the reset signal line **501_n**. A first terminal of the preset transistor **M4** is configured to receive an operation voltage V_{dd} . A second terminal of the preset transistor **M4** is electrically connected to the driving transistor **M2** and the reset transistor **M3**. A control terminal of the preset transistor **M4** is configured to receive a preset signal $Prst(n)$ from the preset signal line **503_n**. An anode of the sensing unit **PD** is electrically connected to a bias voltage V_{bias} . A cathode of the sensing unit **PD** is electrically connected to the driving transistor **M2** and the reset transistor **M3** through the circuit node **N1**.

[0031] In the embodiment of the disclosure, the readout circuit **621** in the peripheral circuit **520** includes a reset control transistor **Trer**, a readout transistor **Tro**, an amplifier circuit **621_1** and a

current source **621_2**. A first terminal of the reset control transistor Trer is configured to receive a reset voltage Vrst from the reset voltage line **506**. A second terminal of the reset control transistor Trer is electrically connected to the circuit node N2 and the readout transistor Tro. A control terminal of the reset control transistor Trer is configured to receive a reset control signal Rst_ctrl from the reset control line **505**. A first terminal of the readout transistor Tro is electrically connected to the circuit node N2 and the reset control transistor Trer. A second terminal of the readout transistor Tro is electrically connected to the amplifier circuit **621_1** and the current source **621_2** through a circuit node N3. A control terminal of the readout transistor Tro is configured to receive a readout signal RO from the readout control line **507**. An input terminal of the amplifier circuit **621_1** is electrically connected to the readout transistor Tro and the current source **621_2** through the circuit node N3. The current source **621_2** is electrically connected between the circuit node N3 and a ground or a power supply, and configured to generate a constant current Ics flowing from the circuit node N3 to the ground or a power supply.

[0032] In the embodiment of the disclosure, the read transistor M1, the driving transistor M2, the reset transistor M3, the preset transistor M4, the reset control transistor Trer and the readout transistor Tro may be N-type transistors, but the disclosure is not limited thereto. In the embodiment of the disclosure, the first terminal and the second terminal of the above transistors may be a drain terminal and a source terminal respectively, and the control terminal of the transistor may be a gate terminal, but the disclosure is not limited thereto. In one embodiment of the disclosure, at least one of the read transistor M1, the driving transistor M2, the reset transistor M3, the preset transistor M4, the reset control transistor Trer and the readout transistor Tro may be a P-type transistor. In the embodiment of the disclosure, the sensing unit PD may be a photodiode, but the disclosure is also not limited thereto.

[0033] FIG. 7 is a schematic diagram of voltage variation according to the embodiment of FIG. 6 of the disclosure. Referring to FIG. 4 to FIG. 7, the pixel unit **611** and the readout circuit **621** of FIG. 6 may perform the following steps S410 to S440. In step S410, the preset transistor M4 may be turned on, and a first voltage may be provided to the control terminal of the driving transistor M2. As shown in FIG. 7, the pixel unit **611** may operate a reset process during a reset period RP from time t0 to time t3, and the reset process includes a preset process during the period from time t0 to time t1. During the period from time t0 to time t1, the reset signal RS(n) and the preset signal Prst(n) may be changed from a low voltage level to a high voltage level respectively, and the read signal RD(n), the readout signal RO and the reset control signal Rst_ctrl may be maintained at the low voltage level. Thus, the reset transistor M3 and the preset transistor M4 may be turned on, and the read transistor M1, the reset control transistor Trer and the readout transistor Tro may be turned off, and the operation voltage Vdd (i.e. the first value) may be provided to the control terminal of the driving transistor M2. The preset transistor M4 is turned on during the reset process. More specifically, the control terminal and the first terminal of the driving transistor M2 may be short-circuited by the turned on reset transistor M3, and the driving transistor M2 may be operated as an equivalent diode. A cathode terminal of the equivalent diode corresponds to the second terminal (drain terminal) of the driving transistor M2. An anode terminal of the equivalent diode corresponds to the control terminal and the first terminal of the driving transistor M2. Because the Read transistor M1 is turned off, there is no current flowing through the equivalent diode. Therefore, a voltage Vsub.BE (i.e. the first voltage) of the circuit node N1 is equal to the operation voltage Vdd during the period from time t0 to time t2.

[0034] In step S420, the preset transistor M4 may be turned off, and a second voltage may be provided to the driving transistor M2. As shown in FIG. 7, during the period from time t2 to time t3, the preset signal Prst(n) may be changed from the high voltage level to the low voltage level, the reset signal RS(n) may be maintained at the high voltage level, and the read signal RD(n) and the reset control signal Rst_ctrl may be changed from the low voltage level to the high voltage level. Thus, the read transistor M1, the reset transistor M3 and the reset control transistor Trer may be

turned on, and the preset transistor M4 and the readout transistor Tro may be turned off, and the reset voltage Vrst (i.e. the second voltage) may be provided to the driving transistor M2. More specifically, the control terminal and the first terminal of the driving transistor M2 may also be short-circuited by the turned on reset transistor M3, and the driving transistor M2 may also be operated as the equivalent diode. Thus, due to the reset voltage Vrst is smaller than the voltage of the first terminal of the driving transistor M2, the equivalent diode is operated in forward bias, and there is a diode current flowing through the equivalent diode. Therefore, the voltage V.sub.BE (i.e. the first voltage) of the circuit node N1 is pulled down after time t3, and after certain time, the voltage V.sub.BE (i.e. the first voltage) of the circuit node N1 is equal to the voltage (i.e. the second value) of the reset voltage Vrst plus a threshold voltage Vth (i.e. $V_{rst} + V_{th}$) of the driving transistor M2. That is, the voltage V.sub.BE (i.e. the first voltage) of the circuit node N1 is reduced from the operation voltage Vdd (i.e. the first value) to the voltage (i.e. the second value) of the reset voltage Vrst plus a threshold voltage Vth (i.e. $V_{rst} + V_{th}$) when the reset voltage Vrst is provided to the driving transistor M2.

[0035] Moreover, during the period from time t2 to time t5, the voltage Vdata(m) of the circuit node N2 is equal to the reset voltage Vrst. Then, during a data initialization period DIP from time t5 to time t6, the electronic device 500 may operate a data initialization process. During the data initialization period DIP from time t5 to time t6, the reset signal RS(n), the preset signal Prst(n), the read signal RD(n) and the reset control signal Rst_ctrl may be maintained at the low voltage level respectively, and the readout signal RO may be changed from the low voltage level to the high voltage level. Thus, the read transistor M1, the reset transistor M3, the preset transistor M4 and the reset control transistor Trer may be turned off, and the readout transistor Tro may be turned on, and the voltage Vdata(m) of the circuit node N2 is pulled down from the reset voltage Vrst to a low voltage level (i.e. ground voltage).

[0036] In step S430, the pixel unit 611 executes a sensing process. As shown in FIG. 7, the electronic device 500 may operate the sensing process during an exposure period EP from time t7 to time t8, and the sensing unit PD may provide a sensing signal (sensing current) corresponding to a sensing result to the control terminal of the driving transistor M2. During the exposure period EP from time t7 to time t8, the reset signal RS(n), the preset signal Prst(n), the read signal RD(n) and the readout signal RO may be maintained at the low voltage level. Thus, the voltage V.sub.BE (i.e. the first voltage) of the circuit node N1 is further reduced from the voltage (i.e. the second value) of the reset voltage Vrst plus a threshold voltage Vth (i.e. $V_{rst} + V_{th}$) to the voltage (i.e. the third value) of the reset voltage Vrst minus a sensing voltage $\Delta V_{sub.BE}$ and plus a threshold voltage Vth (i.e. $V_{rst} - \Delta V_{sub.BE} + V_{th}$). The sensing voltage $\Delta V_{sub.BE}$ represents a voltage variation due to the sensing result of the sensing unit PD.

[0037] In step S440, the pixel unit 611 executes a scan process to provide an output signal. As shown in FIG. 7, the pixel unit 611 may operate a scan process during a scan period SP from time t9 to time t12. During the scan period SP from time t9 to time t12, the preset signal Prst(n) may be changed from the low voltage level to the high voltage level to turn on the preset transistor M4. During a readout period ROP from time t10 to time t11, the read signal RD(n) and the readout signal RO may be changed from the low voltage level to the high voltage level respectively, and the reset signal RS(n), and the reset control signal Rst_ctrl may be maintained at the low voltage level. Thus, the read transistor M1, the preset transistor M4 and the readout transistor Tro may be turned on, and the reset transistor M3 and the reset control transistor Trer may be turned off, and the readout circuit 621 may readout the output signal with the voltage Vdata(m) through the circuit node N2 according to the voltage V.sub.BE (i.e. the first voltage) of the circuit node N1. More specifically, the constant current Ics is flowing through the preset transistor M4, the driving transistor M2, the read transistor M1 and the readout transistor Tro and toward the current source 621_2. The driving transistor M2 may act as a source follower in the saturation region, and the voltage Vdata(m) of the circuit node N2 is pulled up from the low voltage level (i.e. ground

voltage) to the voltage of the reset voltage V_{rst} minus the sensing voltage $\Delta V_{sub.BE}$ and minus a transistor voltage β (i.e. $(V_{rst} - \Delta V_{sub.BE} + V_{th}) - V_{th} - \beta = V_{rst} - \Delta V_{sub.BE} - \beta$). That is, the threshold voltage V_{th} can be excluded in the voltage $V_{data(m)}$. The transistor voltage β may be expressed in the above equation (1).

[0038] Therefore, the input terminal of the amplifier circuit **621_1** may receive an input voltage V_{in} from the circuit node **N3**, and the input voltage V_{in} is equal to the voltage $V_{data(m)}$ (i.e. $V_{rst} - \Delta V_{sub.BE} - \beta$). That is, the readout circuit **621** may generate the sensing data that is not affected by the threshold voltage V_{th} of the driving transistor **M2**.

[0039] In summary, the electronic device and the operating method thereof of the disclosure can effectively eliminate or reduce the threshold voltage (V_{th}) problem of the driving transistor, so as to effectively improve the sensing quality of image sensing results.

[0040] It will be apparent to those skilled in the art that various modifications and variations can be made to the disclosed embodiments without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the disclosure covers modifications and variations provided that they fall within the scope of the following claims and their equivalents.

Claims

1. An electronic device, comprising: a driving transistor, having a control terminal and a first terminal; a reset transistor, electrically connected to the control terminal of the driving transistor; a sensing unit, electrically connected to the control terminal of the driving transistor; and an amplifier circuit, having an input terminal, wherein the first terminal of the driving transistor is electrically connected to the input terminal of the amplifier circuit, wherein a voltage value corresponding to the control terminal of the driving transistor is reduced from a first value to a second value during the reset transistor is turned on.
2. The electronic device according to claim 1, further comprising: a current source, electrically connected to the input terminal of the amplifier circuit.
3. The electronic device according to claim 1, further comprising: a read transistor, electrically connected to the input terminal of the amplifier circuit.
4. The electronic device according to claim 3, wherein the read transistor is electrically connected to the input terminal of the amplifier circuit through a readout transistor.
5. The electronic device according to claim 4, wherein the readout transistor and the input terminal of the amplifier circuit are electrically connected to a current source.
6. The electronic device according to claim 1, wherein the sensing unit is a photodiode.
7. The electronic device according to claim 1, wherein the sensing unit has an anode terminal electrically connected to a bias voltage, and a cathode terminal electrically connected to the control terminal of the driving transistor.
8. The electronic device according to claim 1, wherein the sensing unit is electrically connected between the control terminal of the driving transistor and a bias voltage.
9. The electronic device according to claim 1, wherein the reset transistor is electrically connected between the control terminal of the driving transistor and the first terminal of the driving transistor.
10. The electronic device according to claim 1, wherein the driving transistor has a second terminal, and the second terminal of the driving transistor is electrically connected to an adjustable voltage.
11. The electronic device according to claim 1, further comprising: a preset transistor, electrically connected between the input terminal of the amplifier circuit and a preset voltage.
12. The electronic device according to claim 11, wherein the driving transistor is located in an active area of the electronic device, and the preset transistor is located in a peripheral area of the electronic device.
13. The electronic device according to claim 1, wherein the driving transistor has a second

terminal, and the reset transistor is electrically connected between the control terminal of the driving transistor and second terminal of the driving transistor.

14. The electronic device according to claim 13, further comprising: a preset transistor, electrically connected to the second terminal of the driving transistor and an operation voltage.

15. The electronic device according to claim 14, wherein the driving transistor and the preset transistor are located in an active area of the electronic device.

16. The electronic device according to claim 1, wherein the first terminal of the driving transistor is electrically connected to the input terminal of the amplifier circuit through a read transistor.
